

Introduction

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Abstract

This project develops a multiple linear regression model to estimate used vehicle selling prices from vehicle characteristics, condition, mileage, model, and geographic location. The resulting model explains approximately 82% of the variation in observed selling prices and provides a framework for supporting vehicle acquisition, pricing, and inventory decisions.

Keywords: Regression, Used Vehicles, Predictive Analytics, Vehicle Valuation

The objective of this project is to develop a predictive model capable of estimating the market value of used vehicles. Accurate vehicle valuation supports pricing decisions for dealerships, fleet operators, auction participants, and vehicle purchasing organizations.

The business decision addressed by this analysis is:

Given a used vehicle's characteristics, what selling price should be expected in the current market?

The resulting model can be used to identify vehicles that appear underpriced or overpriced relative to market expectations.

1 Business Objective

The primary objective is to minimize pricing error between predicted and observed vehicle sale prices.

A practical loss function for this problem is Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where:

- y_i is the observed selling price
- $\hat{y}_i$ is the predicted selling price

Lower prediction error improves decision quality by reducing the likelihood of overpaying for inventory or undervaluing vehicles during resale.

2 Plan

The analysis follows the regression modeling workflow:

1. Define the business problem.
2. Acquire and clean vehicle sales data.
3. Explore relationships between candidate predictors and selling price.
4. Build an interpretable regression model.
5. Evaluate assumptions and model fit.
6. Generate predictions and business recommendations.

The ideal dataset would contain:

- Vehicle age
- Odometer reading
- Mechanical condition
- Original MSRP
- Vehicle trim level
- Accident history
- Maintenance history
- Regional market conditions
- Vehicle Make/Model
- Sell price

Several of these variables were unavailable. Vehicle model and state were therefore used as proxy variables for differences in vehicle desirability and regional pricing effects.

3 Build

3.1 Data Source

The dataset was obtained from the Kaggle used vehicle auction dataset used throughout this project.

Variables initially available included:

- Year
- Make
- Model
- Transmission
- State
- Condition
- Odometer
- Selling price

3.2 Data Cleaning

The following preprocessing steps were performed:

- Removed records with missing values in critical fields.
- Calculated vehicle age from model year.
- Converted odometer values into units of 10,000 miles.
- Standardized categorical values.
- Grouped low-frequency vehicle models into manufacturer-specific “Other” categories.
- Grouped low-frequency states into broader state categories.
- Created dummy variables for categorical predictors.

3.3 Sampling Strategy

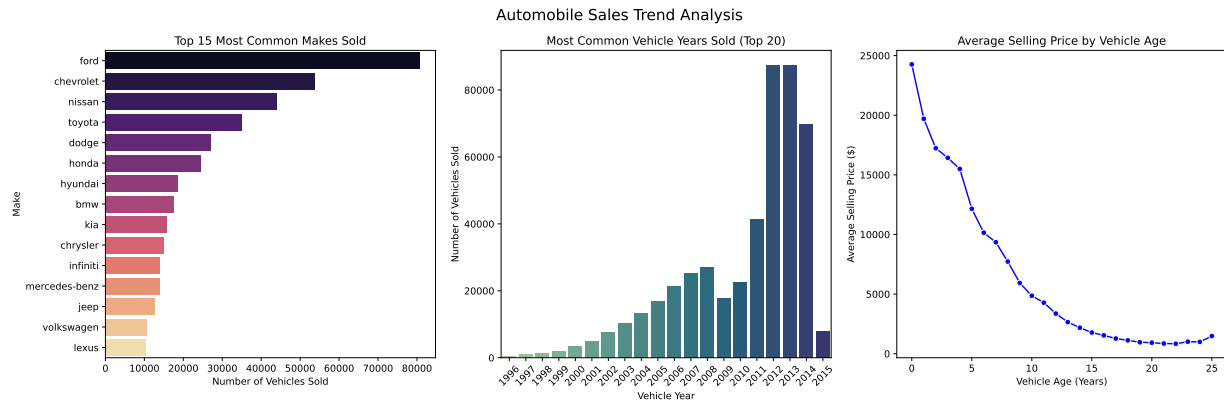
A stratified sampling approach was used to preserve state representation while reducing dataset size to 5000 for modeling.

Training and testing partitions were generated after sampling to maintain representative geographic distributions.

4 Explore

4.1 Exploratory Data Analysis

Initial visualizations focused on the relationships between selling price and major numerical predictors.



- A log transformation was applied to the selling price.

4.2 Vehicle Age vs Selling Price

Vehicle age showed a strong negative relationship with price. Older vehicles generally sold for substantially less than newer vehicles.



4.3 Odometer vs Selling Price

Vehicles with higher mileage generally sold for lower prices. The relationship was weaker than vehicle age but remained statistically significant.



4.4 Key Findings

- Vehicle age exhibits the strongest continuous predictor effect.
- Mileage has a meaningful negative effect on vehicle value.
- Vehicle model accounts for substantial differences in selling price.
- Trucks and heavy-duty pickups tend to command premium prices.
- Geographic location contributes measurable price variation.

5 Reconcile

5.1 Assumptions

Multiple linear regression requires:

- Linear relationships

- Independent observations
- Constant variance
- Approximately normal residuals
- Limited multicollinearity

5.2 Model Refinements

Several iterations of the model were evaluated.

Adjustments included:

- Consolidating sparse model categories.
- Grouping low-frequency states.
- Transforming odometer readings to improve interpretability.
- Evaluating Variance Inflation Factors (VIFs).

These modifications reduced model complexity while preserving predictive performance.

	Variable	VIF
0	const	178.803946
102	state_grouped_fl	6.907212
100	state_grouped_ca	6.474079
63	model_chevrolet_Other	6.085027
9	model_Altima	4.928018
..	...	...
67	model_fiat_Other	1.193466
90	model_saab_Other	1.107573
93	model_smart_Other	1.103110

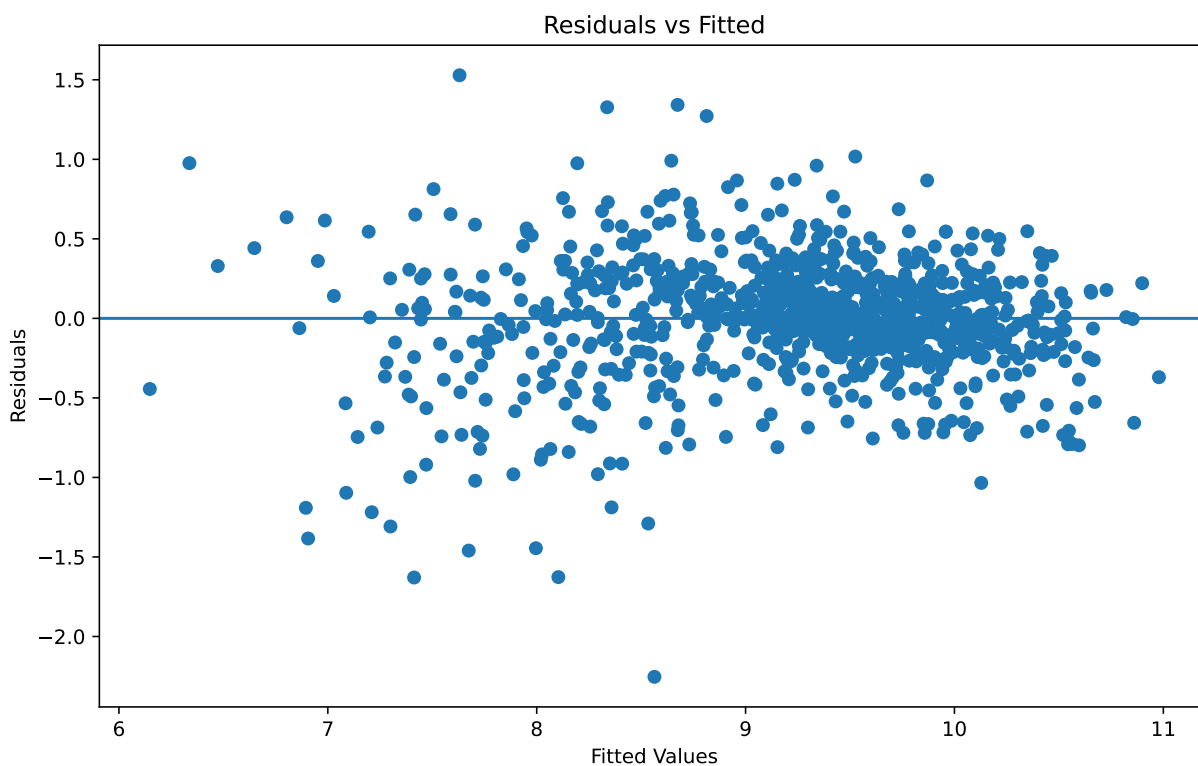

```
74      model_isuzu_Other      1.082286
65      model_daewoo_Other     1.040533
```

```
[124 rows x 2 columns]
```

Our original VIF showed high values with mmr, make, and body. Mmr is a market estimate for the worth of the vehicle so it made sense that predictor overpowered all of the other predictors. Likewise, it's a pretty safe assumption that the vehicle model was already capturing the make and body. For example a vehicle model of 'Ram 1500' only refers to a dodge truck so the model can easily deduce that data from the vehicle model so including those in the model was redundant.

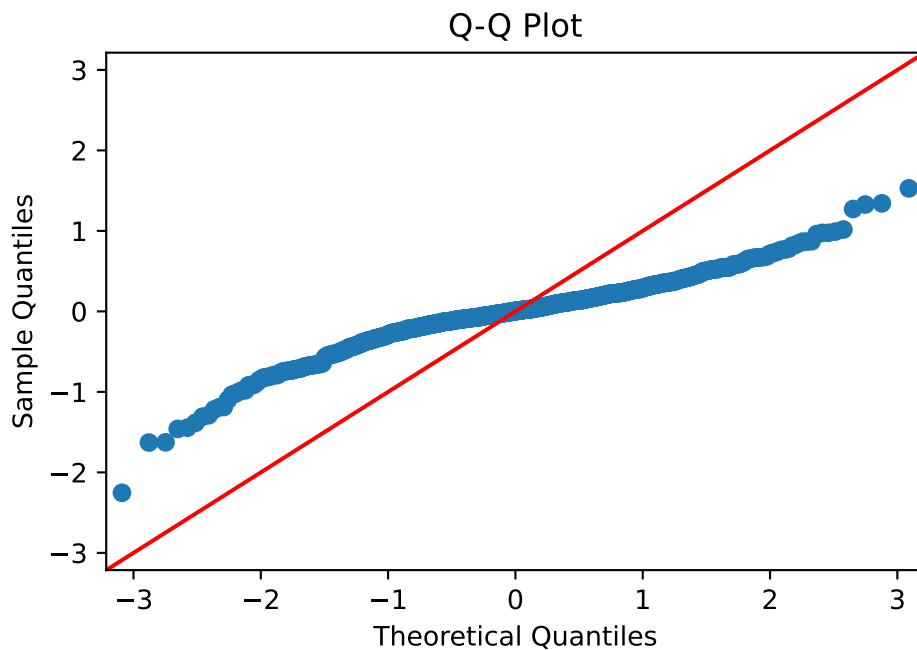
Our current model shows no VIF values above or close to 10 so there appear to be no concerns there.

```
Text(0.5, 1.0, 'Residuals vs Fitted')
```



This graph shows a decent cloud not indicating any concerns.

```
Text(0.5, 1.0, 'Q-Q Plot')
```



Our QQ plot is not great, but we already applied a log transform so there's not much more that can be done. This QQ plot shape is not uncommon with sales data and some of the shape may be accounted for that we have vehicle model and state predictors that are categorical rather than linear and are a significant part of the model.

6 Fit

6.1 Final Model

The final model included:

- Vehicle age
- Odometer (10,000-mile units)
- Condition score

- Vehicle model
- State group

OLS Regression Results

```

=====
Dep. Variable:          sellingprice  R-squared:          0.818
Model:                  OLS          Adj. R-squared:       0.812
Method:                 Least Squares  F-statistic:        141.6
Date:                  Fri, 19 Jun 2026  Prob (F-statistic):    0.00
Time:                  22:24:48      Log-Likelihood:      -
1715.4
No. Observations:      4000          AIC:                  3679.
Df Residuals:          3876          BIC:                  4459.
Df Model:              123
Covariance Type:       nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975
-----
const          10.0407      0.080     125.821      0.000      9.884
vehicle_age     -0.1279      0.003    -44.877      0.000      -
0.134      -0.122
odometer_10k    -0.0508      0.002    -24.010      0.000      -
0.055      -0.047
condition        0.0073      0.001     14.577      0.000      0.006
model_200       -0.6800      0.097     -6.995      0.000      -

```

0.871	-0.489					
model_3 Series		0.0911	0.082	1.113	0.266	-
0.069	0.252					
model_300		-0.0872	0.100	-0.876	0.381	-
0.282	0.108					
model_5 Series		0.3066	0.099	3.104	0.002	0.113
model_Accord		-0.2591	0.082	-3.175	0.002	-
0.419	-0.099					
model_Altima		-0.4687	0.075	-6.227	0.000	-
0.616	-0.321					
model_Avenger		-0.7208	0.091	-7.921	0.000	-
0.899	-0.542					
model_C-Class		0.0999	0.090	1.116	0.264	-
0.076	0.275					
model_CR-V		0.1006	0.107	0.941	0.347	-
0.109	0.310					
model_Camry		-0.3019	0.078	-3.865	0.000	-
0.455	-0.149					
model_Charger		-0.0663	0.098	-0.673	0.501	-
0.259	0.127					
model_Civic		-0.4153	0.086	-4.853	0.000	-
0.583	-0.248					
model_Corolla		-0.4822	0.084	-5.756	0.000	-
0.646	-0.318					

model_Cruze	-0.6591	0.090	-7.338	0.000	-
0.835	-0.483				
model_E-Class	0.3532	0.096	3.673	0.000	0.165
model_Edge	-0.0024	0.090	-0.027	0.978	-
0.179	0.174				
model_Elantra	-0.6615	0.086	-7.688	0.000	-
0.830	-0.493				
model_Equinox	-0.2747	0.087	-3.153	0.002	-
0.446	-0.104				
model_Escape	-0.2795	0.077	-3.647	0.000	-
0.430	-0.129				
model_Expedition	0.1691	0.109	1.558	0.119	-
0.044	0.382				
model_Explorer	0.0166	0.085	0.195	0.845	-
0.150	0.184				
model_F-150	0.2495	0.078	3.181	0.001	0.096
model_F-250 Super Duty	0.5993	0.100	5.968	0.000	0.402
model_Focus	-0.7664	0.078	-9.873	0.000	-
0.919	-0.614				
model_Fusion	-0.4933	0.077	-6.412	0.000	-
0.644	-0.342				
model_G Sedan	-0.0277	0.084	-0.331	0.740	-
0.192	0.136				
model_Grand Caravan	-0.3861	0.086	-4.490	0.000	-

0.555	-0.218					
model_Grand Cherokee		0.0804	0.097	0.832	0.406	-
0.109	0.270					
model_Impala		-0.5320	0.084	-6.326	0.000	-
0.697	-0.367					
model_Jetta		-0.5260	0.100	-5.246	0.000	-
0.723	-0.329					
model_Journey		-0.3347	0.095	-3.531	0.000	-
0.521	-0.149					
model_Liberty		-0.1972	0.109	-1.817	0.069	-
0.410	0.016					
model_Malibu		-0.5346	0.085	-6.321	0.000	-
0.700	-0.369					
model_Maxima		-0.2231	0.087	-2.569	0.010	-
0.393	-0.053					
model_Mazda3		-0.5291	0.099	-5.332	0.000	-
0.724	-0.335					
model_Mustang		-0.2246	0.103	-2.187	0.029	-
0.426	-0.023					
model_Odyssey		-0.0549	0.098	-0.562	0.574	-
0.247	0.137					
model_Optima		-0.4698	0.094	-4.988	0.000	-
0.654	-0.285					
model_Passat		-0.6219	0.094	-6.632	0.000	-

0.806	-0.438					
model_RAV4		-0.0871	0.104	-0.836	0.403	-
0.291	0.117					
model_Rogue		-0.3026	0.096	-3.151	0.002	-
0.491	-0.114					
model_Sentra		-0.6501	0.090	-7.223	0.000	-
0.827	-0.474					
model_Sienna		0.1062	0.107	0.991	0.322	-
0.104	0.316					
model_Silverado 1500		0.2541	0.087	2.932	0.003	0.084
model_Sonata		-0.5277	0.088	-6.028	0.000	-
0.699	-0.356					
model_Sorento		-0.3940	0.103	-3.843	0.000	-
0.595	-0.193					
model_Soul		-0.6376	0.105	-6.050	0.000	-
0.844	-0.431					
model_Suburban		0.4482	0.104	4.309	0.000	0.244
model_Tahoe		0.5444	0.100	5.462	0.000	0.349
model_Taurus		-0.6687	0.105	-6.360	0.000	-
0.875	-0.463					
model_Town and Country		-0.3667	0.093	-3.946	0.000	-
0.549	-0.184					
model_Versa		-0.7535	0.108	-6.969	0.000	-
0.965	-0.542					

model_Wrangler	0.4013	0.092	4.343	0.000	0.220
model_X5	0.4439	0.108	4.094	0.000	0.231
model_acura_Other	0.1363	0.089	1.535	0.125	-
0.038 0.310					
model_audi_Other	0.2036	0.088	2.318	0.021	0.031
model_bmw_Other	0.3983	0.087	4.569	0.000	0.227
model_buick_Other	-0.3924	0.093	-4.213	0.000	-
0.575 -0.210					
model_cadillac_Other	-0.0152	0.081	-0.186	0.852	-
0.175 0.145					
model_chevrolet_Other	-0.2908	0.073	-3.962	0.000	-
0.435 -0.147					
model_chrysler_Other	-1.1530	0.099	-11.659	0.000	-
1.347 -0.959					
model_daewoo_Other	-2.2839	0.385	-5.931	0.000	-
3.039 -1.529					
model_dodge_Other	-0.1590	0.080	-1.997	0.046	-
0.315 -0.003					
model_fiat_Other	-0.9297	0.168	-5.519	0.000	-
1.260 -0.599					
model_ford_Other	-0.2788	0.078	-3.562	0.000	-
0.432 -0.125					
model_gmc_Other	0.2408	0.083	2.918	0.004	0.079
model_honda_Other	-0.0278	0.097	-0.287	0.774	-

0.217	0.162					
model_hummer_Other		0.8827	0.144	6.140	0.000	0.601
model_hyundai_Other		-0.4713	0.084	-5.607	0.000	-
0.636	-0.306					
model_infiniti_Other		0.2024	0.084	2.407	0.016	0.038
model_isuzu_Other		-0.9221	0.278	-3.320	0.001	-
1.467	-0.378					
model_jaguar_Other		0.2494	0.129	1.935	0.053	-
0.003	0.502					
model_jeep_Other		-0.3907	0.090	-4.343	0.000	-
0.567	-0.214					
model_kia_Other		-0.7087	0.083	-8.545	0.000	-
0.871	-0.546					
model_land rover_Other		0.6535	0.129	5.075	0.000	0.401
model_lexus_Other		0.3985	0.080	4.990	0.000	0.242
model_lincoln_Other		-0.1519	0.093	-1.631	0.103	-
0.334	0.031					
model_mazda_Other		-0.5526	0.089	-6.230	0.000	-
0.727	-0.379					
model_mercedes-benz_Other		0.5828	0.088	6.641	0.000	0.411
model_mercury_Other		-0.4185	0.138	-3.042	0.002	-
0.688	-0.149					
model_mini_Other		-0.3755	0.107	-3.515	0.000	-
0.585	-0.166					

model_mitsubishi_Other	-0.6594	0.094	-6.988	0.000	-
0.844	-0.474				
model_nissan_Other	-0.1042	0.078	-1.333	0.183	-
0.258	0.049				
model_pontiac_Other	-0.7952	0.097	-8.183	0.000	-
0.986	-0.605				
model_porsche_Other	1.0412	0.150	6.931	0.000	0.747
model_ram_Other	0.3523	0.158	2.224	0.026	0.042
model_saab_Other	-0.7784	0.229	-3.393	0.001	-
1.228	-0.329				
model_saturn_Other	-0.7523	0.101	-7.435	0.000	-
0.951	-0.554				
model_scion_Other	-0.5546	0.150	-3.701	0.000	-
0.848	-0.261				
model_smart_Other	-1.1332	0.229	-4.949	0.000	-
1.582	-0.684				
model_subaru_Other	-0.1688	0.096	-1.750	0.080	-
0.358	0.020				
model_suzuki_Other	-0.6913	0.138	-5.004	0.000	-
0.962	-0.420				
model_toyota_Other	0.1528	0.077	1.994	0.046	0.003
model_volkswagen_Other	-0.2980	0.093	-3.221	0.001	-
0.479	-0.117				
model_volvo_Other	-0.2249	0.098	-2.284	0.022	-

0.418	-0.032					
state_grouped_az		0.0218	0.062	0.349	0.727	-
0.100	0.144					
state_grouped_ca		0.1581	0.044	3.599	0.000	0.072
state_grouped_co		0.0942	0.067	1.399	0.162	-
0.038	0.226					
state_grouped_fl		0.1232	0.043	2.842	0.005	0.038
state_grouped_ga		0.1410	0.047	3.004	0.003	0.049
state_grouped_il		0.1443	0.050	2.909	0.004	0.047
state_grouped_in		0.0447	0.078	0.574	0.566	-
0.108	0.197					
state_grouped_ma		-0.0426	0.066	-0.648	0.517	-
0.171	0.086					
state_grouped_md		-0.0296	0.059	-0.503	0.615	-
0.145	0.086					
state_grouped_mi		0.1618	0.054	3.024	0.003	0.057
state_grouped_mn		0.2002	0.060	3.332	0.001	0.082
state_grouped_mo		0.1418	0.053	2.678	0.007	0.038
state_grouped_nc		0.1157	0.051	2.268	0.023	0.016
state_grouped_ne		0.1334	0.080	1.676	0.094	-
0.023	0.290					
state_grouped_nj		0.0773	0.049	1.565	0.118	-
0.020	0.174					
state_grouped_nv		0.1024	0.056	1.836	0.066	-

0.007	0.212					
state_grouped_ny		0.0479	0.072	0.661	0.508	-
0.094	0.190					
state_grouped_oh		0.0613	0.050	1.225	0.221	-
0.037	0.159					
state_grouped_pa		0.1496	0.049	3.054	0.002	0.054
state_grouped_sc		0.1564	0.076	2.059	0.040	0.007
state_grouped_tn		0.1271	0.050	2.528	0.011	0.029
state_grouped_tx		0.0745	0.045	1.646	0.100	-
0.014	0.163					
state_grouped_va		0.0361	0.059	0.617	0.537	-
0.079	0.151					
state_grouped_wa		0.0933	0.064	1.462	0.144	-
0.032	0.218					
state_grouped_wi		0.1179	0.059	1.989	0.047	0.002

=====					
Omnibus:	499.223	Durbin-Watson:		2.020	
Prob(Omnibus):	0.000	Jarque-Bera (JB):		3013.924	
Skew:	-0.431	Prob(JB):		0.00	
Kurtosis:	7.164	Cond. No.		3.85e+03	
=====					

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.85e+03. This might indicate that there are strong multicollinearity or other numerical problems.

The fitted model produced:

- $R^2 = 0.818$
- Adjusted $R^2 = 0.812$
- F-statistic = 141.6
- $p < 0.001$

indicating strong overall explanatory power.

6.2 Important Continuous Predictors

Variable	Coefficient
Vehicle Age	-0.1279
Odometer (10k miles)	-0.0508
Condition	0.0073

Because the dependent variable is log-transformed, coefficients can be approximately interpreted as percentage changes.

6.2.1 Vehicle Age

Each additional year reduces expected selling price by approximately:

$$100(e^{-0.1279} - 1) \approx -12.0\%$$

6.2.2 Odometer

Each additional 10,000 miles reduces expected selling price by approximately:

$$100(e^{-0.0508} - 1) \approx -5.0\%$$

6.2.3 Condition

Each one-point increase in condition increases expected selling price by approximately:

$$100(e^{0.0073} - 1) \approx 0.7\%$$

7 Evaluate

7.1 Residual Analysis

Residual diagnostics indicated:

- No severe violations of linearity.
- Acceptable residual symmetry.
- Some heavy tails remain.
- No extreme multicollinearity concerns.

The largest VIF values were associated with popular truck categories but remained below commonly cited thresholds requiring corrective action.

7.2 Limitations

Several limitations should be noted:

- No MSRP information was available.
- Trim level information was not incorporated.
- Accident history was unavailable.
- Maintenance records were unavailable.
- Auction data may not represent all vehicle markets.

These limitations likely explain a portion of the unexplained variance.

8 Predict

8.1 Business Implications

The model suggests:

1. Vehicle age is the strongest continuous driver of depreciation.
2. Mileage has a significant but smaller effect.
3. Vehicle model contributes substantial value differences.
4. Trucks generally retain value better than passenger sedans.

8.2 Recommendations

Organizations purchasing used vehicles should:

- Prioritize lower-age vehicles when possible.
- Consider mileage alongside age rather than in isolation.
- Use model-specific adjustments when evaluating value.
- Compare observed asking prices against model predictions to identify unusually favorable purchasing opportunities.

8.3 Future Work

Future improvements could include:

1. Incorporating original MSRP.
2. Including trim-level information.
3. Adding accident-history variables.
4. Incorporating local economic indicators.

9 Conclusion

The final regression model explains approximately 82% of the variation in used vehicle selling prices and provides a practical framework for vehicle valuation. Vehicle age, mileage, model, and geographic location all contribute significantly to price determination. The resulting model can support data-driven purchasing and pricing decisions while highlighting areas for future model enhancement.

References